FIELD JOURNAL 0014

Model sequence dependence with Markov chains

How can a model predict an ordered process without remembering its entire history?



Event totals tell us what happens, but not in what order. A Markov chain uses observed sequences to predict what comes next.

I UNDERSTAND THE PROBLEM *What are we trying to predict?*

1 Ordered behavior matters

Shoppers move through a website in a sequence. Event totals alone miss the order, which is essential for predicting what comes next.

Given the current event, what is the most likely next event?

We use ten observed shopping journeys as a running example. Section 2 lists these journeys as the data used throughout the infographic.

2 Observed shopping journeys (data)

Ten complete sessions (J1 occurs twice).
Purchase = bought, Exit = left without buying.

J1 × 2	Search → View → Compare → Cart → Purchase *(observed twice)*
J2 × 1	Search → View → Compare → Cart → Exit
J3 × 1	Search → View → Cart → Purchase
J4 × 1	Search → View → Cart → Exit
J5 × 1	Search → View → Compare → Exit
J6 × 1	Search → View → Exit
J7 × 1	Search → View → Compare → View → Cart → Purchase
J8 × 1	Search → View → Compare → View → Cart → Exit
J9 × 1	Search → View → Cart → Compare → Cart → Purchase

START / END are implicit session boundaries.
Same evidence is used throughout.

3 Same events, different next steps

Two excerpts from J1 and J9 contain the same events, but in a different order.

J1 excerpt: View → Compare → Cart

J9 excerpt: View → Cart → Compare

Same three events; different current states.
After Cart → Purchase is likeliest (5/9).
After Compare → Cart is likeliest (4/7).

4 Event totals vs. next-event prediction

Event	Total visits (across all journeys)
Search	10
View	12
Compare	7
Cart	9
Purchase	5
Exit	5

These totals tell us how often each event appears, but not what typically happens next.

A Markov chain models the order in order to predict the next event.

II BUILD THE MODEL *How do we model sequence dependence with a Markov chain?*

5 The Markov assumption (first-order)

The next event depends only on the current event, not the entire past.

$$P(X_{t+1} = j \mid X_t = i, X_{t-1}, \ldots, X_1) = P(X_{t+1} = j \mid X_t = i)$$

What the assumption means
Once we know the current state i, earlier states do not add predictive information. The next state j depends only on the current state.

$X_t = i$	current state (e.g., *View*)
$X_{t+1} = j$	next state (e.g., *Cart*)
$X_{t-1}, \ldots, X_1$	earlier states (history)

Example
Two shoppers are both on *View* but arrived by different paths (e.g., from *Search* or from *Compare*). A first-order model gives them the same next-step probabilities.

We also assume time-homogeneous transitions: the probabilities do not change over time.

6 Estimate transition probabilities from data

We estimate each transition probability using observed counts.

6a Count transitions
T_{ij} = number of times event i is followed by event j in the data (adjacent events).

6b Normalize each row

$$\hat{p}_{ij} = \frac{T_{ij}}{\sum_k T_{ik}}$$

Divide each count by the total number of departures from i (each row sums to 1).

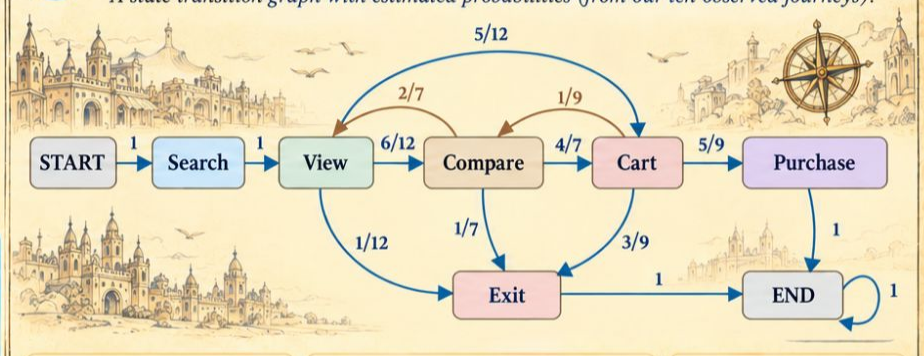
Example: View → Compare
In our ten observed journeys, *View* is followed by *Compare* 6 times, and there are 12 total departures from *View*.

$T_{\text{View,Compare}} = 6$ (View → Compare transitions) ÷ $\sum_k T_{\text{View},k} = 12$ (Total departures from View) = $\hat{P}_{\text{View,Compare}} = 0.50$ (50%)

Maximum likelihood estimation (MLE)
These observed proportions are the MLEs: among all possible transition probabilities, they assign the highest likelihood to the observed transition counts, given the model's structure.

7 Complete first-order Markov model for the shopping journey

A state transition graph with estimated probabilities (from our ten observed journeys).



Arrows show the probability of moving to the next event, conditional on the current event. Each state's outgoing probabilities sum to 1.0.

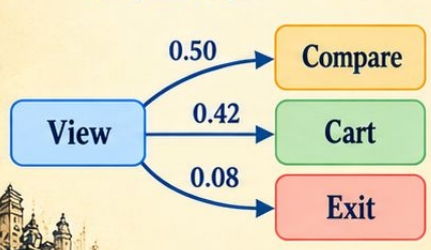
Return paths include Compare → View and Cart → Compare. Purchase and Exit each lead to END (probability 1). Only END is absorbing.

This graph and the transition matrix (next section) are two representations of the same estimated transition probabilities.

III USE THE MODEL *What can we predict or calculate with a Markov chain?*

8 One-step prediction

From a current state, the model gives a probability distribution over the next state. This tells us the most likely next step(s).



Compare is the most likely next step, but Cart and Exit are also possible. The model predicts a distribution, not a certainty.

9 Multi-step prediction (two routes)

We can use the model to find the probability of reaching *Exit* in exactly two steps from *View*, by considering all possible routes.

Multiply along each route, then add across routes.

Route 1: View → Compare → Exit
$0.50 \times 1/7 = 0.071$

Route 2: View → Cart → Exit
$5/12 \times 3/9 = 0.139$

Total probability: 21.03%

$$P(X_{t+2} = \textit{Exit} \mid X_t = \textit{View}) = 0.071 + 0.139 = 0.210$$

10 Generate a shopping journey

Here is one possible shopper journey sampled from the fitted Markov chain, starting at *Search* and continuing step by step until *END*.

Search → View → Compare → Cart → Purchase → END

Sampling does not always choose the most likely edge. Different simulations can produce different journeys.

11 Matrix powers: multi-step reachability

Matrix powers summarize many possible multi-step paths at once. P^k gives the k-step transition probabilities between all pairs of states.

$$P^k = P \times P \times \cdots \times P \quad (k \text{ times})$$

$$(P^2)_{ij} = \sum_m P_{im} P_{mj}$$

For every possible intermediate state m, multiply the two step probabilities and add the results.

Section 9 is one entry of P^2: it adds all two-step routes from *View* to *Exit*.

Larger k looks farther into the future.

12 Eigenvalues: what survives repeated transitions?

The two eigenvectors of the transition matrix identify the stationary distribution and the shrinking deviation, letting us compute the distribution after k steps.

1 Start with a two-state chain

Consider a simple chain with two states, A and B, and transition matrix:

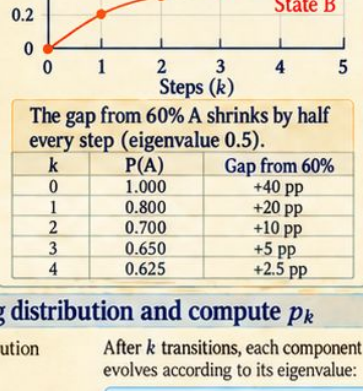
$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}$$

- From A: 80% stay in A, 20% move to B.
- From B: 30% move to A, 70% stay in B.
- Start in State A:

$p_0 = (1, 0)$.

2 Repeated transitions converge to a stable distribution

Probability moves both ways between A and B. Starting from everyone in A, the distribution approaches 60% A / 40% B.



The gap from 60% A shrinks by half every step (eigenvalue 0.5).

k	P(A)	Gap from 60%
0	1.000	+40 pp
1	0.800	+20 pp
2	0.700	+10 pp
3	0.650	+5 pp
4	0.625	+2.5 pp

3 The two eigenpairs of P

Every finite transition matrix has an eigenvalue of 1, because the rows sum to 1. The corresponding left eigenvector is a stationary distribution. Our matrix P has a second eigenvalue 0.5, which identifies a direction of deviation from stationarity.

For a left eigenvector v and eigenvalue λ: $vP = \lambda v$.

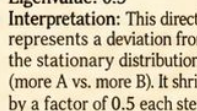
Eigenpair 1: eigenvalue 1 (stationary distribution)
$(0.6, 0.4)P = 1 \cdot (0.6, 0.4)$

- Left eigenvector: (0.6, 0.4)
- Eigenvalue: 1
- **Interpretation:** This is the stationary distribution. It remains the same after every transition.

Stationary distribution
60% A / 40% B

Eigenpair 2: eigenvalue 0.5 (deviation from stationarity)
$(1, -1)P = 0.5 \cdot (1, -1)$

- Left eigenvector: (1, −1)
- Eigenvalue: 0.5
- **Interpretation:** This direction represents a deviation from the stationary distribution (more A vs. more B). It shrinks by a factor of 0.5 each step.



Deviation shrinks by half each step

4 Decompose the starting distribution and compute p_k

For this matrix, any starting distribution can be written as a combination of its two eigenvectors.

$(1, 0) = (0.6, 0.4) + (0.4, -0.4)$

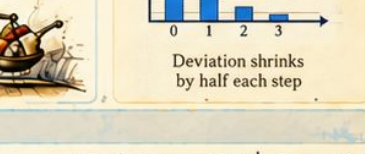
Stationary component (stays the same) · Deviation component (shrinks by 0.5^k)

After k transitions, each component evolves according to its eigenvalue:

$p_k = p_0 P^k = (0.6, 0.4) + (0.5)^k (0.4, -0.4)$

The stationary component persists (eigenvalue 1). The deviation component shrinks by 0.5^k.

After k transitions, $(0.5)^k$ of the initial deviation remains, so the distribution moves toward 60% A / 40% B.



Each step multiplies the deviation by 0.5.

Key takeaway

- Every finite row-stochastic matrix has eigenvalue 1; a stationary distribution is a left eigenvector.
- Other eigenvalues describe deviations. Here 0.5 means the gap halves after each transition.
- After k steps, $p_k = (0.6, 0.4) + (0.5)^k(0.4, -0.4)$: one component stays; the other fades.
- Convergence is not automatic. This chain reaches 60/40; the shopping chain reaches absorbing END.

13 Likelihood of a complete journey

The probability of a complete shopping journey is the product of the transition probabilities along the path.

1 An observed shopping journey (J1)

Return to our ten shopping journeys. J1 follows Search → View → Compare → Cart → Purchase.
What probability does our fitted model assign to this complete route?

Search → View → Compare → Cart → Purchase

1.00 (1/1)	0.50 (6/12)	0.57 (4/7)	0.56 (5/9)

2 Compute the journey probability

Multiply the transition probabilities along the path:

$$P(J1) = 1 \times \frac{6}{12} \times \frac{4}{7} \times \frac{5}{9} = \frac{10}{63} \approx 0.1587$$

15.87% model-implied probability

3 Use a log likelihood

Logs turn products of small probabilities into sums of logarithms. This simplifies long-sequence scoring and helps prevent numerical underflow.

$$\ln P(J1) = \ln(10/63) \approx -1.8405$$

4 Model vs. observed frequency

J1 appeared in 2 of 10 observed journeys (20%).
The fitted model assigns it a probability of 15.87%.

Journey	Observed frequency	Model-implied probability
J1	2/10 = 20.00%	15.87%

These are not the same. The model estimates each transition using evidence pooled across all journeys, not just this one.

Key takeaway

- Journey likelihood multiplies the transition probabilities along its path.
- Log likelihood adds log probabilities and helps prevent underflow.
- J1's 15.87% uses pooled model evidence; observed frequency is 20%.

IV CHALLENGE THE MODEL *What are the limitations, and how can we improve the model?*

14 Zero probabilities

Some transitions never appear in the training data, giving a probability of zero.

Example: an unseen but possible transition
$P(\textit{Purchase} \mid \textit{Compare}) = 0$

- None of our 10 training journeys includes *Compare → Purchase*, so its estimated probability is zero.
- Any journey containing this transition also receives zero probability.
- **Unseen does not mean impossible.** Smoothing can assign probability to eligible transitions.

15 Smoothing (add-one / Laplace)

Add a small count to every eligible transition to avoid zero probabilities.

$$\hat{p}_{ij} = \frac{T_{ij} + \alpha}{\sum_k T_{ik} + \alpha K}$$

- $\alpha = 1$ is add-one (Laplace) smoothing.
- K = number of eligible next events (here $K = 4$).

Example: next event after Compare ($K = 4$)

Next event	Count T_{ij}	MLE ($\alpha = 0$)	Add-one ($\alpha = 1$)
View	2	2/7 = 28.6%	3/11 = 27.3%
Cart	4	4/7 = 57.1%	5/11 = 45.5%
Purchase	0	0/7 = 0%	1/11 = 9.1%
Exit	1	1/7 = 14.3%	2/11 = 18.2%

Seven observed departures + four pseudocounts = 11.
Smoothing **redistributes probability** without creating observations. Forbidden transitions do not receive pseudocounts.

16 Second-order model (remember two events)

Instead of just the current event, remember the previous event as well. A state is a pair (previous, current).

How the state updates

(Search, View) → (View, Compare) → (Compare, Cart)

At each step, drop the oldest event and add the newest.

Evidence is split across contexts

Our 10 training journeys contain 12 departures from *View*. A second-order model splits these into:

- 10 after Search → View
- 2 after Compare → View

Example transition distributions (raw MLE)

From (Search, View) (10 departures)

Next event	Count	Probability
Compare	6	6/10 = 0.60
Cart	3	3/10 = 0.30
Exit	1	1/10 = 0.10

From (Compare, View) (2 departures)

Next event	Count	Probability
Compare	0	0/2 = 0
Cart	2	2/2 = 1.00
Exit	0	0/2 = 0

The cost of more context

With N event types, an order-k model can have up to N^k contexts and $N^k(N-1)$ free parameters. More memory divides the evidence.

100	10,000	1,000,000
first order contexts	second order contexts	third order contests

Note: The 100% estimate above is based on only 2 training examples. Smoothing reduces extremes; more history splits evidence.

17 Evaluation with held-out data

Use unseen journeys to compare models. A better model assigns higher likelihood to new data.

Train on 10 observed journeys (estimate P) → Test on 2 unseen journeys (compute log likelihood) → Compare models

Worked example: Test journey A

Held-out journey: *Search → View → Compare → View → Compare → Cart → Purchase*
Search → View has probability 1 and is omitted, so five transitions are shown below.

First-order model (current state only)
$P_1(\textit{Test A}) = (7/15)(3/11)(7/15)(5/11)(6/12)$
$\ln P_1 \approx -4.3052$

Second-order model (previous and current)
$P_2(\textit{Test A}) = (7/13)(3/10)(1/5)(4/10)(4/7)$
$\ln P_2 \approx -4.9084$

Key difference: the third factor. First order pools all *View* departures (7/15), while second order uses the specific context (*Compare, View*) (1/5).

Log-likelihood scores (higher is better)

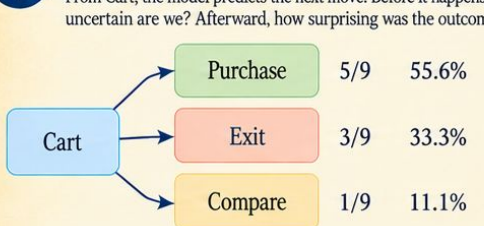
Model	Test journey A	Test journey B	Combined
First-order	−4.3052	−4.4128	−8.7180
Second-order	−4.9084	−4.1744	−9.0827

First order scores higher across these two held-out journeys.
More memory does not guarantee better predictions, especially with limited evidence.

V INTERPRET THE MODEL *What does probability tell us about surprise and uncertainty?*

18 Surprisal and entropy (same next-event distribution)

From Cart, the model predicts the next move. Before it happens, how uncertain are we? Afterward, how surprising was the outcome?



Nine observed departures from Cart: five purchases, three exits, and one return to Compare.

Surprisal: "How surprising is one event?"

Surprisal measures how unexpected an event was after it occurred. Less likely events produce more surprise.

$$I(j) = -\log_2 P(j \mid \text{current state})$$

Next event j	Probability $P(j \mid \text{Cart})$	Surprisal $I(j)$ (bits)
Purchase	5/9 = 0.556	0.85
Exit	3/9 = 0.333	1.58
Compare	1/9 = 0.111	3.17

Purchase (5/9) produces 0.85 bits; Compare (1/9) produces 3.17 bits. More surprising doesn't mean mistaken.

Entropy: "What is the average uncertainty?"

Entropy is the probability-weighted average surprisal before we know the next event.

$$H(\text{Cart}) = -\sum_j P(j \mid \text{Cart}) \log_2 P(j \mid \text{Cart})$$
$$= \frac{5}{9}(0.85) + \frac{3}{9}(1.58) + \frac{1}{9}(3.17)$$
$$\approx \mathbf{1.35 \text{ bits}}$$

Our Cart distribution **1.35 bits** — Purchase is more likely than the other outcomes, so the next event is somewhat predictable.

Three equally likely outcomes **1.58 bits** — Maximum entropy for 3 outcomes ($\log_2 3$). More uncertainty if all next events are equally likely.

Entropy describes uncertainty about the next event — not uncertainty in our estimated probabilities.

19 Field Note Model assumptions and checks

When applying a Markov chain to a new problem, check:

1. **States.** Choose observable, meaningful states with clear interpretations.
2. **Eligible transitions.** Assign probability only to transitions that can actually occur.
3. **Session boundaries.** Use START and END states to mark complete journeys.
4. **Time stability.** Check whether transition probabilities remain stable over time.
5. **Smoothing.** Handle unseen but eligible transitions with smoothing.
6. **Model order.** Choose memory length (first order, second order, etc.) using held-out evaluation.
7. **Interpretation.** Fitted probabilities do not establish shopper intent or causation. A high probability does not mean a step is desirable or will inevitably occur.

Whole-journey surprise: connecting to likelihood

A journey's surprise is the sum of its transition surprises, because logarithms turn probability products into sums.

$P(\text{journey}) = \frac{10}{63} \approx 15.87\%$ → $\text{Surprise(journey)} = -\log_2 \frac{10}{63} \approx \mathbf{2.66 \text{ bits}}$

This is the same journey probability we calculated earlier (Chapter 13), now expressed as total surprise in bits.

- More surprising journeys have lower likelihood under the model.
- In Chapter 17 we used natural logarithms (ln) to compute log-likelihood scores in nats. Here we use base-2 logarithms ($\log_2$) measured in bits. They are the same idea on different scales.

Remember: what Markov chains enable

A Markov chain learns transition probabilities from observed journeys to predict future states, score journeys, and quantify uncertainty. More history may improve predictions, but divides the evidence across more contexts.

Observed journeys (training data) → Transition model (transition probabilities) → Predictions, likelihoods, uncertainty (new data)

Sources: Univ. of Michigan MADS (Data Mining II): Jurafsky & Martin, Speech and Language Processing (2026 draft); Aggarwal, Data Mining: The Textbook, Ch. 15